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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 213 — Past Questions & Materials  
**by Renance**

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Course: PHY 213  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 9  
Levels: 200, 300

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This study pack was compiled and restructured by Renance for PHY 213 students at Federal University of Petroleum Resources, Effurun. It brings together 9 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

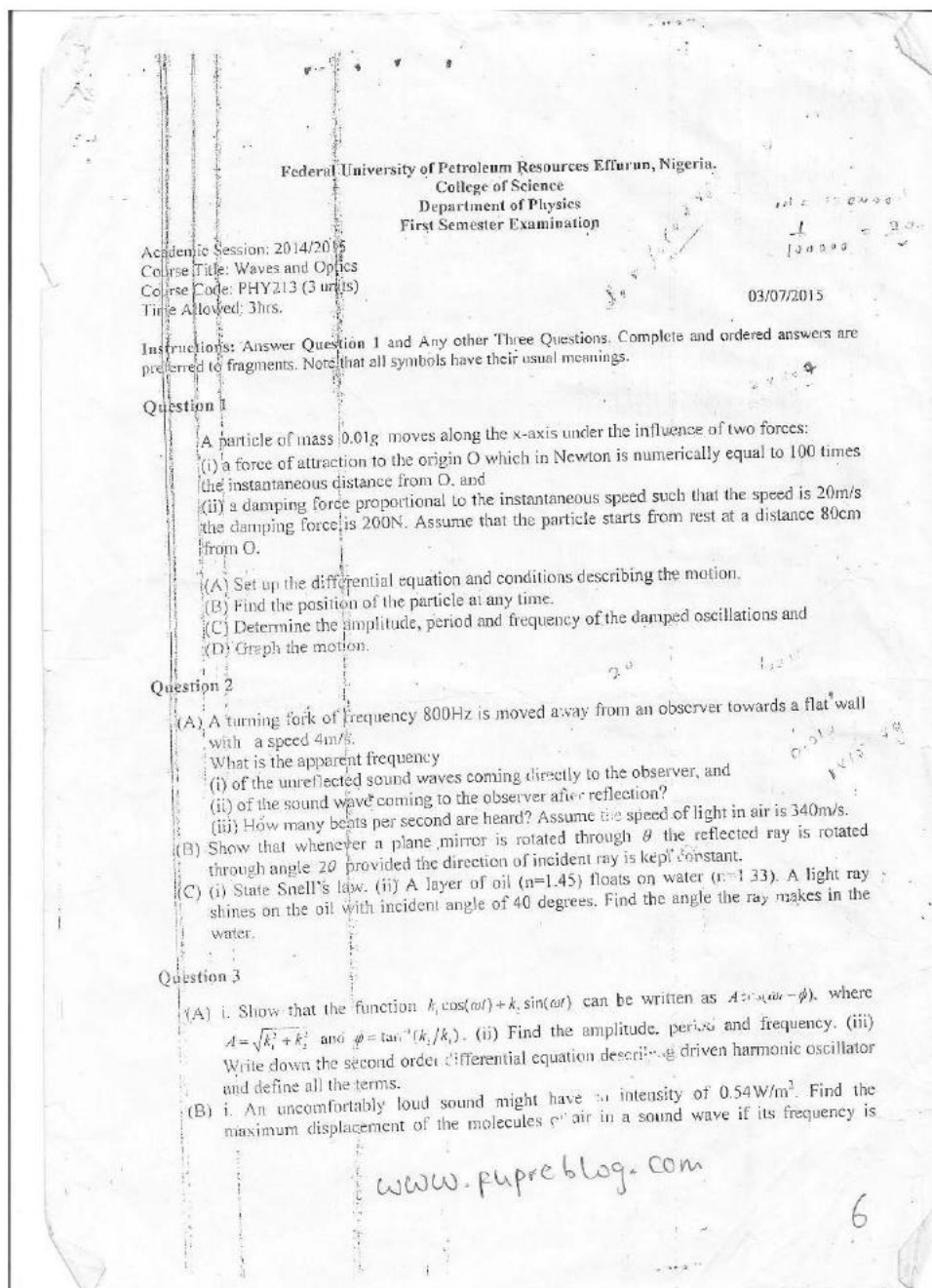
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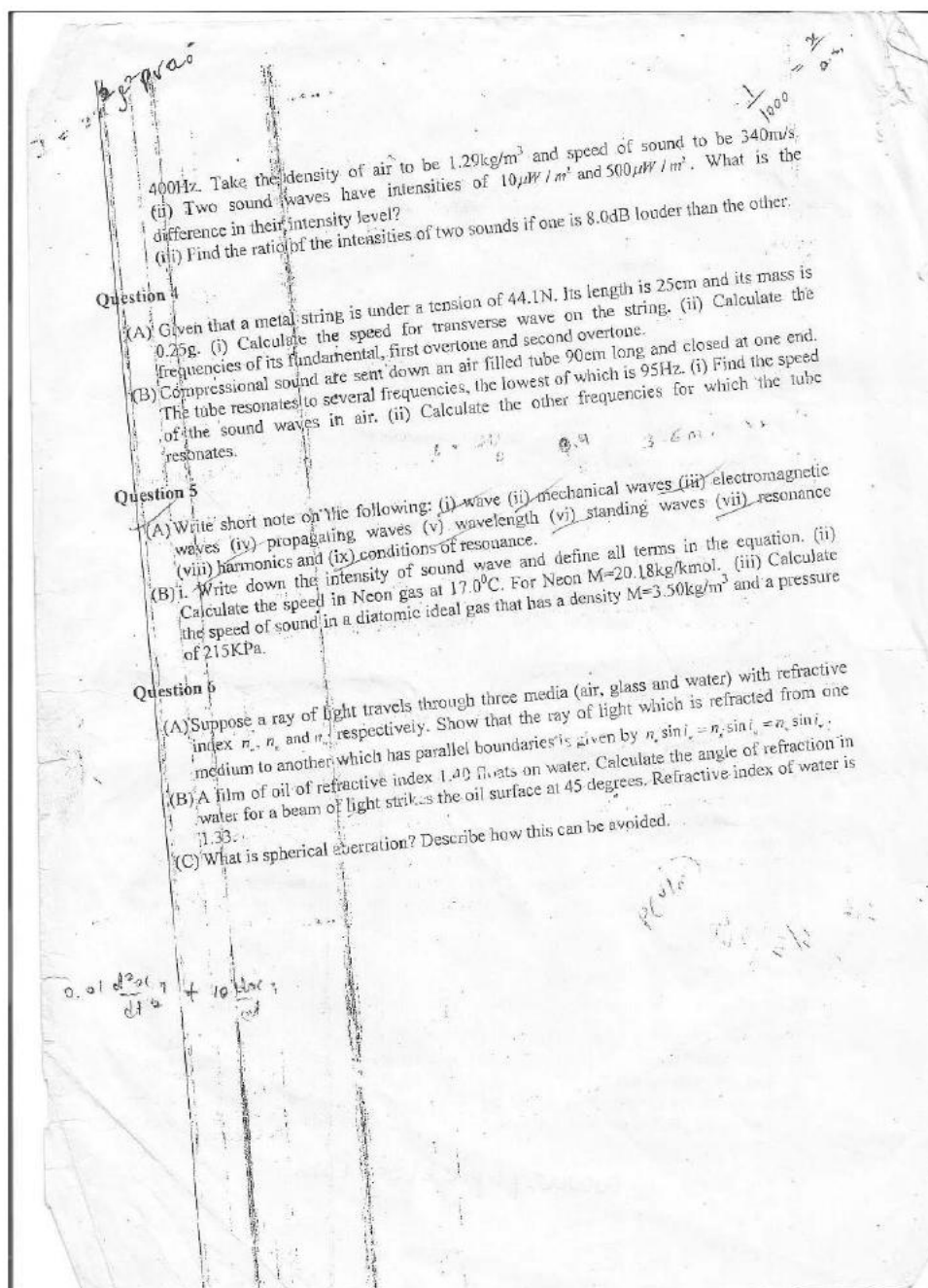
# Paper 1 of 9 — PHY 213 - 2014/2015 - First Semester - Waves and Optics

Level 300 • First Semester • PastQuestion

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## Paper 2 of 9 — PHY 213 - 2015/2016 - First Semester - Waves and Optics

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION

Academic Session: 2015/2016

Course Title: Waves and Optics

Course Code: PHY 213 (3 Units)

Time Allowed: 2hrs 30mins Date: 06/05/2016

Instructions: Answer any four (4) questions but two (2) questions must be answered from each section

### SECTION A (WAVES)

#### Question 1

- a) When does Simple Harmonic Motion occur?
- b) (i) What do you understand by forced vibration  
(ii) With the aid of diagrams explain 3 types of damped vibration
- c) State the Principle of Superposition
- d) A train of simple harmonic wave travelling in a gas along the positive direction of the x-axis with an amplitude 2cm, velocity 45m/s and frequency 75 Hz. Calculate (i) the displacement of the wave (ii) particle velocity (iii) particle acceleration at a distance 135cm from the origin after an interval of 3secs.

#### Question 2

- a) (i) What is Doppler Effect  
(ii) State two applications of Doppler effect
- b) (i) State 5 properties of waves  
(ii) Give three differences between progressive wave and stationary wave
- c) A truck is approaching a road crossing with a speed of 60km/hr, a police man standing near the crossing hears the frequency of the car's horn as 360Hz. What is the actual frequency of the horn (speed of sound in air is 332m/s).
- d) A simple progressive wave is represented by the equation  $y = 2.5 \sin (125t - 31.4x)$  where y and x are expressed in meters and t in seconds. Find (i) Amplitude (ii) wavelength (iii) speed (iv) frequency

#### Question 3

- a) Write short notes on the following (i) Beats (ii) Wavelength (iii) Interference (iv) Diffraction (v) longitudinal wave
- b) State two uses of Beats
- c) Calculate the velocity of sound in a gas in which two waves of length 50cm and 50.4cm produce 6 beats per sec

d) A wire of density  $9 \times 10^3 \text{ kg/m}^3$  is stretched between 2 clamps 1 meter apart while subjected to an extension of 0.05cm. what is the lowest frequency of transverse vibrations in wire assuming Young's modulus of the material to be  $9 \times 10^{10} \text{ N/m}^2$

#### SECTION B (OPTICS)

##### Question 4

- a) In order to obtain a stationary interference pattern, state the three conditions for interference of light
- b) What do you understand by critical angle
- c) (i) State the laws of reflection  
(ii) With the aid of diagrams differentiate between concave and convex mirror
- d) In a young slit experiment, the separation between the first and fifth bright fringe is 2.5mm when the wavelength used is  $6.2 \times 10^{-7} \text{ m}$ . If the distance from the slits to the screen is 0.8m. Calculate the separation of the 2 slits in mm.

##### Question 5

- a) (i) Define diffraction grating  
(ii) Define Dispersive Power of Grating  
(iii) State two types of grating of light wave
- b) (i) State two conditions for Total Internal Reflection to occur  
(ii) State the laws of Refraction
- c) If the object distance from concave mirror is 20cm and its focal length is 10cm.  
Find (i) the image distance  
(ii) is the image real or virtual
- d) In a Newton's Ring experiment that has the radius of curvature of the plano convex lens to be 100cm, if the diameter of the 5th Ring was 0.3cm and diameter of the 25th Ring was 0.8cm, what is the value of the wavelength of light used.

##### Question 6

- a) Explain in details <Newton's Ring Experiment=>
- b) List three types of Fringes
- c) The refractive index of glass with respect to water is 1.125, if the absolute refractive index of glass is 1.5. find the absolute refractive index of water
- d) A Biprism is placed 5cm from a slit illuminated by sodium light of wavelength  $5890 \times 10^{-8} \text{ cm}$ . if the width of the fringes obtained on a screen 75cm from the biprism is  $9.424 \times 10^{-2} \text{ cm}$ . What is the distance between the two coherent sources

# Paper 3 of 9 — PHY 213 - 2016/2017 - First Semester - Waves and Optics

Level 200 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.  
College of Science, Department of Physics  
First Semester Examination

Academic Session: 2016/2017  
Course Title: Waves and Optics  
Course Code: PHY213 (3 units)

Time: 2hrs.30mins.  
Date: 30/03/2017

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

**Question 1**

A particle of mass 10kg moves along the x-axis under the influence of two forces:

- a force of attraction to the origin O which in Newton is numerically equal to 80 times the instantaneous distance from O, and
- a damping force proportional to the instantaneous speed such that when the speed is 10m/s the damping force is 400N. Assuming the particle starts from rest at a distance 40m from O.

- Set up the differential equation and conditions describing the motion.
- Find the position of the particle at any time.
- Determine the amplitude, period and frequency of the damped oscillations and
- Sketch the motion of this damped oscillation.

(22 marks)

**Question 2**

- In a Newton's Ring experiment the diameter of the 5<sup>th</sup> ring was 0.3cm and the diameter of 25<sup>th</sup> ring was 0.008m. If the wavelength of the light is  $6.87 \times 10^{-5}$  cm, calculate the radius of curvature of the plano-convex lens used for the experiment.
  - With a well labeled diagram, explain Newton's ring experiment.
- A parallel beam of sodium light is allowed to be incident normally on a plane grating having 5042 lines/cm and a second order spectral line is observed to be deviated through 38°. Calculate the wavelength of spectral line.
  - Define Dispersive power of grating and Dispersion of light

(16 marks)

**Question 3**

- The separation between the second and sixth bright fringe is 2.5mm when the wavelength used is  $6.2 \times 10^{-7}$  m. Calculate the separation of the two slits in mm, if distance between the slits and the screen is 0.8m.
  - If the two straight and narrow parallel slits 1mm apart are illuminated by monochromatic light, the fringes formed on the screen held at a distance of 100cm from slits are 0.5mm apart. What is the wavelength of the monochromatic light?
- A layer of oil ( $n=1.45$ ) floats on water ( $n=1.33$ ). A light ray shines on the oil with incident angle of 40 degrees. Find the angle the ray makes in the water.
  - State Snell's law.
  - Explain the two types of interference of light

(16 marks)

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## Question 4

- (A) (i) Explain the term "HOLOGRAPHY" and its two steps processes.  
 (ii) With a well labeled diagram, explain how a hologram can be recorded and viewed  
 (iii) List five applications of holography.
- (B) Show that whenever a plane mirror is rotated through  $\theta$  the reflected ray is rotated through angle  $2\theta$  provided the direction of incident ray is kept constant.
- (C) A tuning fork of frequency 400Hz is moved away from an observer towards a flat wall with a speed 2m/s. What is the apparent frequency of the  
 (i) un-reflected sound waves coming directly to the observer, and  
 (ii) sound wave coming to the observer after reflection?  
 (iii) How many beats per second are heard? Assume the speed of light in air is 340m/s.

(16 marks)

## Question 5

- (A) (i) Show that the function  $k_1 \cos(ax) + k_2 \sin(ax)$  can be written as  $A \cos(ax - \phi)$ , where  $A = \sqrt{k_1^2 + k_2^2}$  and  $\phi = \tan^{-1}(k_2/k_1)$ . (ii) Find the amplitude, period and frequency.  
 (iii) Write down the second order differential equation describing driven harmonic oscillator and define all the terms.
- (B) (i) An uncomfortably loud sound might have an intensity of  $0.54 \text{ W/m}^2$ . Find the maximum displacement of the molecules of air in a sound wave if its frequency is 400Hz. Take the density of air to be  $1.29 \text{ kg/m}^3$  and speed of sound to be  $340 \text{ m/s}$ . (ii) Two sound waves have intensities of  $10 \mu\text{W/m}^2$  and  $500 \mu\text{W/m}^2$ . What is the difference in their intensity level? (iii) Find the ratio of the intensities of two sounds if one is 8.0dB louder than the other.

(16 marks)

## Question 6

- (A) Write short note on the following: (i) wave (ii) mechanical waves (iii) electromagnetic waves (iv) propagating waves (v) wavelength (vi) standing waves (vii) resonance and (viii) harmonics.
- (B) (i) Write down the intensity of sound wave and define all terms in the equation. (ii) Calculate the speed in Neon gas at  $17.0^\circ\text{C}$ . For Neon  $M=20.18 \text{ kg/kmol}$ . (iii) Calculate the speed of sound in a diatomic ideal gas that has a density of  $3.50 \text{ kg/m}^3$  and a pressure of  $215 \text{ KPa}$ .

(16 marks)

## Paper 4 of 9 — PHY 213 - 2013/2014 - First Semester - Waves and Optics

Level 300 • First Semester • PastQuestion

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ee Federal University of Petroleum Resources Effurun, Nigeria. F

College of Science

Department of Physics

First Semester Examination

Academic Session: 2013/2014 Aer

Course Title: Waves and Optics M,

Course Code: PHY213 (3 units) i .

Time: 2hrs. Date: 18-07-2014 ae

Instructions: Answers to Question 1 and any other three should be submitted. Complete and ordered answers are preferred to fragments. i j

~ Question 1.

\* — A particle of mass 10kg moves along the x-axis under the influence of two forces:

- (i) a force of attraction to the origin O which in Newtons is numerically equal to 80 times the instantaneous distance from O, and
- (ii) a damping force proportional to the instantaneous speed such that the speed is 10m/s the damping force is 400Newtons. Assuming that the particle starts from rest at a distance 40m from O,
- (a) Set up the differential equation and conditions describing the motion.
- (b) Find the position of the particle at any time.
- (c) Determine the 'amplitude, pete and frequency of the damped oscillations and
- (d) graph the motion.

Question 2.

7 (A) A turning fork of frequency 400Hz is moved away from an observer towards a flat wall with a speed of 2.0m/s . What is the apparent frequency

- (i) of the unreflected sound waves coming directly to the observer, and Ce
- (ii) of the sound waves coming to the observer after reflection.
- (iii) How many beats per second are heard? Assume the speed of sound i in air is 340m/s.

(B) Show that whenever a plane mirror is rotated through  $\theta$  the reflected ray is rotated through angle  $2\theta$  provided the direction of the incident ray is kept constant."

(C)L. State Snell's law.

II. A layer of oil ( $n=1.45$ ) floats on water ( $n=1.33$ ), A ray of light shines onto the oil with an incidence angle of 40 degrees. Find the angle the ray makes in the water.

Question 3. '

(A) Suppose a ray of light travels through three media (air, glass and water) with

refractive index  $n_1$ ,  $n_2$ , and  $n_3$ , respectively, Show that the ray of light which is refracted from one medium to another which has parallel boundaries is given by  $n_1 \sin(i_1) = n_2 \sin(i_2) = n_3 \sin(i_3)$  (eg

(B) A film of oil of refractive index 1.40 floats on water. Calculate the angle of, WwW - PPT eb og hl

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refraction in water for a beam of light that strikes the oil surface at 45 degrees.

- Refractive index of water is 1.33

(C) What is spherical aberration? Describe how this can be avoided.

A Question 4,

{A) Write short notes on any seven of the following

I. wave, II. Mechanical waves, III. Electromagnetic waves, IV. Propagating waves,

V. Wavelength, VI. Standing waves, VII. Resonance, VIII. Harmonics, IX. Conditions of resonance. (eg? b

(B) I. Write down the expression for the intensity of sound wave and define all the terms in the equation. II. Calculate the speed in neon gas at  $17.0^\circ\text{C}$ . For neon

$M = 20.18 \text{ kg/kmol}$ . III. Calculate the speed of sound in a diatomic ideal gas that has a density  $M = 3.50 \text{ kg/m}^3$  and a pressure of  $P = 215 \text{ kPa}$ .

Question 5. "se

(A) I. Given that a metal string is under a tension of  $44.1 \text{ N}$ . Its length is  $25 \text{ cm}$  and its mass is  $0.25 \text{ g}$ . (i) Calculate the speed for transverse wave on the string. (ii) Calculate the frequencies of its fundamental, first overtone and second overtone.

(A) II. Compressional waves (sound waves) are sent down an air filled tube  $90 \text{ cm}$  long and closed at one end. The tube resonates to several frequencies, the lowest of which is  $95 \text{ Hz}$ . (i) Find the speed of sound, waves in air. (ii) Calculate the other frequencies for which the tube resonates. - :

Question 6, J

(A) I. Show that the function  $c \cos(\omega t) + c \sin(\omega t)$  can be written as  $A \cos(\omega t - \phi)$ , where  $A = \sqrt{c^2 + c^2}$  and  $\phi = \tan^{-1}(1)$ . (A) II. Find the amplitude, period and frequency.

1 :

(A) III. Write down the second order differential equation describing driven harmonic oscillator and define all the terms. ; i

(B) I. An uncomfortably loud sound might have an intensity of  $0.54 \text{ W/m}^2$ . Find the maximum displacement of the molecules of air in a sound wave if its frequency is  $800 \text{ Hz}$ . Take the density of air to be  $1.29 \text{ kg/m}^3$  and speed of sound to be  $340 \text{ m/s}$ .

(B) II. Two sound waves have intensities of  $104 \text{ W/m}^2$  and  $500 \text{ n.W/m}^2$ . What is the difference in their intensity level? ;

f (B)ITL. Find the ratio of the intensities of two sounds if one is 8.0dB louder than the other. i

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**Paper 5 of 9 — PHY 213 2016 2017 EXAM**Level 200 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.

College of Science, Department of Physics

First Semester Examination

Academic Session: 2016/2017

Course Title: Waves and Optics Time: 2hrs.30mins.

Course Code: PHY213 (3 units) Date: 30/03/2017

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

**Question 1**

"A particle of mass 10kg moves along the x-axis under the influence of two forces:

(i) a force of attraction to the origin O which in Newton is numerically equal to 80 times the instantaneous distance from O, and (ii) a damping force proportional to the instantaneous speed such that when the speed is 0m/s the damping force is 400N. Assuming the particle starts from rest at a distance 40m from OQ. '

(Ay Set up the differential equation and conditions describing the motion.

AB) Find the position of the particle at any time.

(C) Determine the amplitude, period and frequency of the damped oscillations and

(DB) Sketch the motion of this damped oscillation.

(22 marks)

**Question 2**

(A) (i) In a Newton's Ring experiment the diameter of the 5<sup>th</sup> ring was 0.3cm and the diameter of 25<sup>th</sup> ring was 0.008m. If the wavelength of the light is  $6.87 \times 10^{-5}$  cm, calculate the radius of curvature of the plano-convex lens used for the experiment.

(ii) With a well labeled diagram, explain Newton's ring experiment.

(8) (i) A parallel beam of sodium light is allowed to be incident normally on a plane grating having 5042 lines/cm and a second order spectral line is observed to be deviated through  $38^\circ$ .

Calculate the wavelength of spectral line.

(ii) Define Dispersive power of grating and Dispersion of light

(16 marks)

**Question 3**

AA) (i) The separation between the second and sixth bright fringe is 2.5mm when the wavelength used is  $6.2 \times 10^{-7}$  m. Calculate the separation of the two slits in mm, if distance between the slits and the screen is 0.8m.

Ai) If the two straight and narrow parallel slits imm apart are illuminated by monochromatic light, the fringes formed on the screen held at a distance of 100cm from

slits are 0.5mm apart. What is the wavelength of the monochromatic light?

4B) (i) A layer of oil ( $n=1.45$ ) floats on water ( $n=1.33$ ). A light ray shines on the oil with incident angle of 40 degrees. Find the angle the ray makes in the water.

(ii) State Snell's law.

Aili) Explain the two types of interference of light

(16 marks)

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Question 4

(A) (i) Explain the term "HOLOGRAPHY" and its two steps processes.

(ii) With a well labeled diagram, explain how a hologram can be recorded and viewed

\_fiil) List five applications of holography.

(B) Show that whenever a plane mirror is rotated through  $\theta$  the reflected ray is rotated through angle  $2\theta$  provided the direction of incident ray is kept constant.

(C) A tuning fork of frequency 400Hz is moved away from an observer towards a flat wall with a speed 2m/s. What is the apparent frequency of the

(i) un-reflected sound waves coming directly to the observer, and

(ii) sound wave coming to the observer after reflection?

(iii) How many beats per second are heard? Assume the speed of light in air is 340m/s.

(16 marks)

Question 5

(A) (i) Show that the function  $k \cos(at) + k \sin(wt)$  can be written as  $A \cos t$

$A = \sqrt{k^2 + k^2}$ ; and  $g = \tan^{-1}(k/k)$ . (ii) Find the amplitude, period and frequency.

(iii) Write down the second order differential equation describing driven harmonic oscillator and define all the terms.

(B) (i) An uncomfortably loud sound might have an intensity of  $0.54 \text{ W/m}^2$ . Find the maximum displacement of the molecules of air in a sound wave if its frequency is 400Hz. Take the density of air to be  $1.29 \text{ kg/m}^3$  and speed of sound to be 340m/s. (ii) Two sound waves have intensities of  $10 \text{ g/m}^2$  and  $500 \text{ pW/m}^2$ . What is the difference in their intensity level? (iii) Find the ratio of the intensities of two sounds if one is 8.0dB louder than the other.

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(16 marks)

4 Question 6

A) Write short note on

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the following: 4i) wave ii) mechanical waves- (iii) electromagnetic waves  
) standing waves iii) resonance and  
speed of sound  
 $215\text{KPa}$ .  
(16 marks)

## Paper 6 of 9 — PHY 213 2013 2014 EXAM

Level 200 • First Semester • PastQuestion

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Fe Federal University of Petroleum Resources Effurun, Nigeria.

2 College of Science

Department of Physics

First Semester Examination

Academic Session: 2013/2014 iis

Course Title: Waves and Optics \ is

Course Code: PHY213 (3 units)

Time: hrs. Date: 18-07-2014 wv.

Instructions: Answers to Question 1 and any other three should be submitted. Complete and ordered answers are preferred to fragments. :

— Question 1.

' A particle of mass 10kg moves along the x-axis under the influence of two forces:

- (i) a force of attraction to the origin O which in Newtons is numerically equal to 80 times the instantaneous distance from O, and
- (ii) a damping force proportional to the instantaneous speed such that the speed is 10m/s the damping force is 400Newtons. Assuming that the particle starts from rest at a distance 40m from O,
- (a) Set up the differential equation and conditions describing the motion.
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Question 2.

(A) A turning fork of frequency 400.Hz is moved away from an observer towards a flat wall with aspeed of 2.0m/s . What is the apparent frequency

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- (iii) How many beats per second are heard? Assume the speed of sound in air is 340m/s.

(B) Show that whenever a plane mirror is rotated through @ the teflected ray is rotated through angle 26 provided the direction of the incident ray is kept constant. fe

(C)I. State Snell's law.

IL A layer of oil ( $n=1.45$ ) floats on water ( $n=1.33$ ). A ray of light shines onto the oil with an incidence angle of 40 degrees. Find the angle the ray makes in the water.

Question 3.

(A) Suppose a ray of light travels through three media (air, glass and water) with refractive index  $n$ ,  $n_y$  and  $n$ , respectively. Show that the ray of light which is 5 refracted from one medium to another which has parallel boundaries is given by

$n \sin(i) = n' \sin(r)$

(B) A film of oil of refractive index 1.40 floats on water. Calculate the angle of refraction in water for a beam of light that strikes the oil surface at 45 degrees.

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refraction in water for a beam of light that strikes the oil surface at 45 degrees.

- Refractive index of water is 1.33

(C) What is spherical aberration? Describe how this can be avoided.

Question 4,

(A) Write short notes on any seven of the following

I. Mechanical waves, II. Electromagnetic waves, III. Propagating waves,

IV. Wavelength, V. Standing waves, VI. Resonance, VII. Harmonics, VIII. Conditions of resonance. pene -

(B) I. Write down the expression for the intensity of sound wave and define all the terms in the equation. II. Calculate the speed of sound in neon gas at 17.0°C. For neon

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Question 5. ee

(A) I. Given that a metal string is under a tension of 44.1N. Its length is 25cm and its mass is 0.25g. (i) Calculate the speed for transverse wave on the string. (ii) Calculate the frequencies of its fundamental, first overtone and second overtone.

(A). Compression waves (sound waves) are sent down an air filled tube 90cm long and closed at one end. The tube resonates to several frequencies, the lowest of which is 95Hz. (i) Find the speed of sound waves in air. (ii) Calculate the other frequencies for which the tube resonates. - , ;

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(A) III. Write down the second order differential equation describing driven harmonic oscillator and define all the terms. :

(B) I. An uncomfortably loud sound might have an intensity of  $0.54 \text{ W/m}^2$ . Find the maximum displacement of the molecules of air in a sound wave if its frequency is 1000 Hz. Take the density of air to be  $1.29 \text{ kg/m}^3$  and speed of sound to be  $340 \text{ m/s}$ .

(B) II. Two sound waves have intensities of  $10 \text{ W/m}^2$  and  $500 \mu\text{W/m}^2$ . What is the difference in their intensity level? i :

(B) III. Find the ratio of the intensities of two sounds if one is 8.0dB louder than the other. ,

## Paper 7 of 9 — Waves and Optics PHY 213 PQ

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION

Academic Session: 2015/2016

Course Title: Waves and Optics

Course Code: PHY 213 (3 Units)

Time Allowed: 2hrs 30mins Date: 06/05/2016

Instructions: Answer any four (4) questions but two (2) questions must be answered from each section

### SECTION A (WAVES)

#### Question 1

- a) When does Simple Harmonic Motion occur?
- b) (i) What do you understand by forced vibration  
(ii) With the aid of diagrams explain 3 types of damped vibration
- c) State the Principle of Superposition
- d) A train of simple harmonic wave travelling in a gas along the positive direction of the x-axis with an amplitude 2cm, velocity 45m/s and frequency 75 Hz. Calculate (i) the displacement of the wave (ii) particle velocity (iii) particle acceleration at a distance 135cm from the origin after an interval of 3secs.

#### Question 2

- a) (i) What is Doppler Effect  
(ii) State two applications of Doppler effect
- b) (i) State 5 properties of waves  
(ii) Give three differences between progressive wave and stationary wave
- c) A truck is approaching a road crossing with a speed of 60km/hr, a police man standing near the crossing hears the frequency of the car's horn as 360Hz. What is the actual frequency of the horn (speed of sound in air is 332m/s).
- d) A simple progressive wave is represented by the equation  $y = 2.5 \sin (125t - 31.4x)$  where y and x are expressed in meters and t in seconds. Find (i) Amplitude (ii) wavelength (iii) speed (iv) frequency

#### Question 3

- a) Write short notes on the following (i) Beats (ii) Wavelength (iii) Interference (iv) Diffraction (v) longitudinal wave
- b) State two uses of Beats
- c) Calculate the velocity of sound in a gas in which two waves of length 50cm and 50.4cm produce 6 beats per sec
- d) A wire of density  $9 \times 10^3 \text{ kg/m}^2$  is stretched between 2 clamps 1 meter apart while subjected to

an extension of 0.05cm. what is the lowest frequency of transverse vibrations in wire assuming Young's modulus of the material to be  $9 \times 10^{10} \text{ N/m}^2$

## SECTION B (OPTICS)

### Question 4

- a) In order to obtain a stationary interference pattern, state the three conditions for interference of light
- b) What do you understand by critical angle
- c) (i) State the laws of reflection  
(ii) With the aid of diagrams differentiate between concave and convex mirror
- d) In a young slit experiment, the separation between the first and fifth bright fringe is 2.5mm when the wavelength used is  $6.2 \times 10^{-7} \text{ m}$ . If the distance from the slits to the screen is 0.8m. Calculate the separation of the 2 slits in mm.

### Question 5

- a) (i) Define diffraction grating  
(ii) Define Dispersive Power of Grating  
(iii) State two types of grating of light wave
- b) (i) State two conditions for Total Internal Reflection to occur  
(ii) State the laws of Refraction
- c) If the object distance from concave mirror is 20cm and its focal length is 10cm.  
Find (i) the image distance  
(ii) is the image real or virtual
- d) In a Newton's Ring experiment that has the radius of curvature of the plano convex lens to be 100cm, if the diameter of the 5th Ring was 0.3cm and diameter of the 25th Ring was 0.8cm, what is the value of the wavelength of light used.

### Question 6

- a) Explain in details "Newton's Ring Experiment"
- b) List three types of Fringes
- c) The refractive index of glass with respect to water is 1.125, if the absolute refractive index of glass is 1.5. find the absolute refractive index of water
- d) A Biprism is placed 5cm from a slit illuminated by sodium light of wavelength  $5890 \times 10^{-8} \text{ cm}$ . if the width of the fringes obtained on a screen 75cm from the biprism is  $9.424 \times 10^{-2} \text{ cm}$ . What is the distance between the two coherent sources

## Paper 8 of 9 — PHY 213 2014 2015 EXAM

Level 200 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria. ‘

3 College of Science -

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i Department of Physics Rees

| ; é First Semester Examination s

Ackidentic Session: 2014/20 ES i -t a

Colirse (Ti le: Waves and Optics \* Iyer %

Colirse Code: PHY213 (3 units) me

Tipe Ajlowed: 3hrs. ee a 03/07/2015

In. truc i ‘Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

Lid : ref

Qifstion 1

moves along the x-axis under the influence of two forces:

ij

| <A particle of mass :0.01g

hich in Newton is numerically equal to 100 times

(i) a force of attraction to the origin O w

ithe instantaneous distance from O, and

i(i) a damping force proportional to the instantaneous speed such that the speed is 20m/s

he damping force, is 200N. Assume that the particle starts from rest at a distance 80cm

| ifrom oO. i

i WA) Set up the differential equation and conditions describing the motion.

(B) Find the position of the particle at any time.

! co) Determine the pmlplitude, period and frequency of, the damped oscillations and

: (D) Graph the motion.

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equency 800Hz is moved away from an observer towards a flat wall

(A), A ‘turning fork of |

| with aspeed 4mv/8.

What is the apparent frequency :

; i(1) of the unreflected sound waves coming, directly to the observer, and

(it) of the sound wave coming to the observer after reflection? :

Gi) How many beats per second are heard? Assume the speed of light in air is 340 m/s.

(B)-Show that whenever a plane mirror is rotated through  $\theta$  the reflected ray is rotated through angle  $2\theta$  provided the direction of incident ray is kept constant. .

NC) G) State Snell's law. (ii) A layer of oil ( $n=1.45$ ) floats on water ( $n=1.33$ ). A light ray -  
|” shines on the oil with incident angle of 40 degrees. Find the angle the ray makes in the  
i, water. :

Question 3

that the function  $\cos(wt) + k \sin(wt)$  can be written as  $A \sin(wt + \phi)$ , where  
 $\phi = \tan^{-1}(k/4)$ . (ii) Find the amplitude, period and frequency. (iii)

spring-driven harmonic oscillator

1A) i. Show

$A = \frac{h}{m\lambda}$  and  $\lambda = \frac{c}{f}$

Write down the second order differential equation describing

and define all the terms. : :

(B) i. An uncomfortably loud sound might have intensity of  $0.54 \text{ W/m}^2$ . Find the  
maximum displacement of the molecules of air in a sound wave if its frequency is

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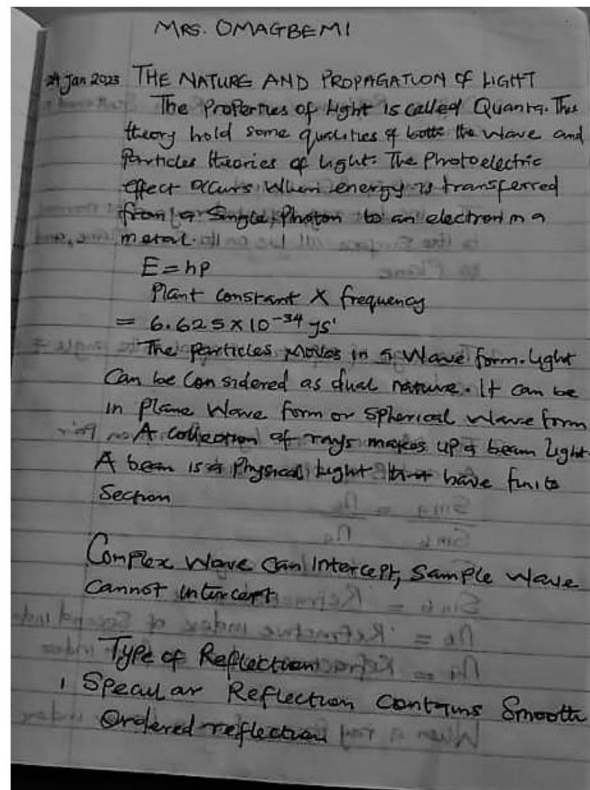
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 M=20.18kg/kmol.  
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 . Calculate the angle  
 Refractive:  
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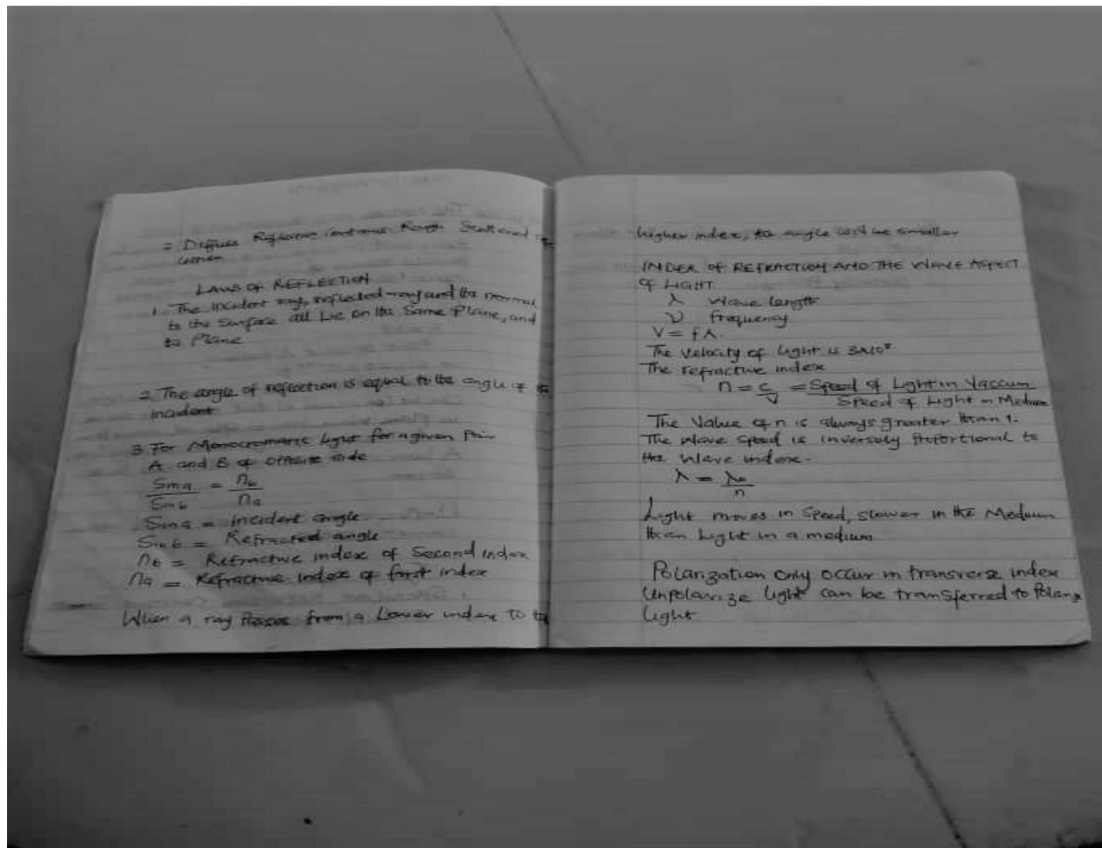
What is the  
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 (i) Find the speed  
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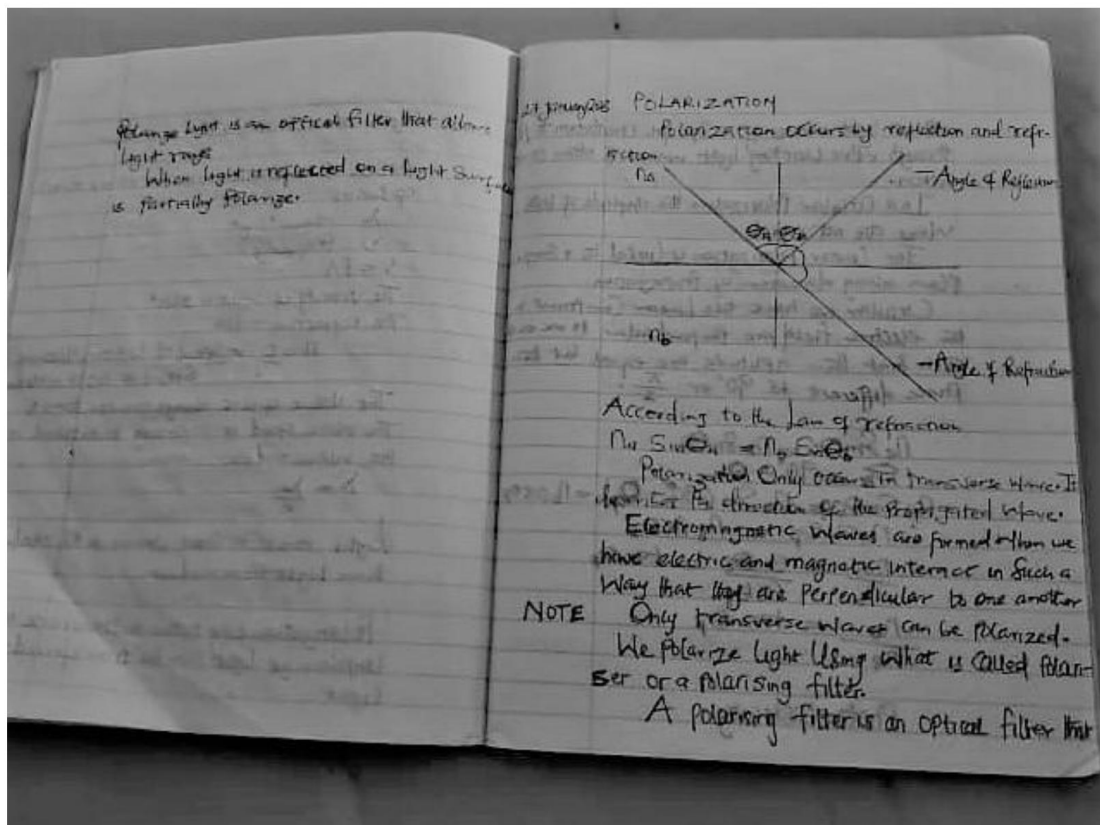
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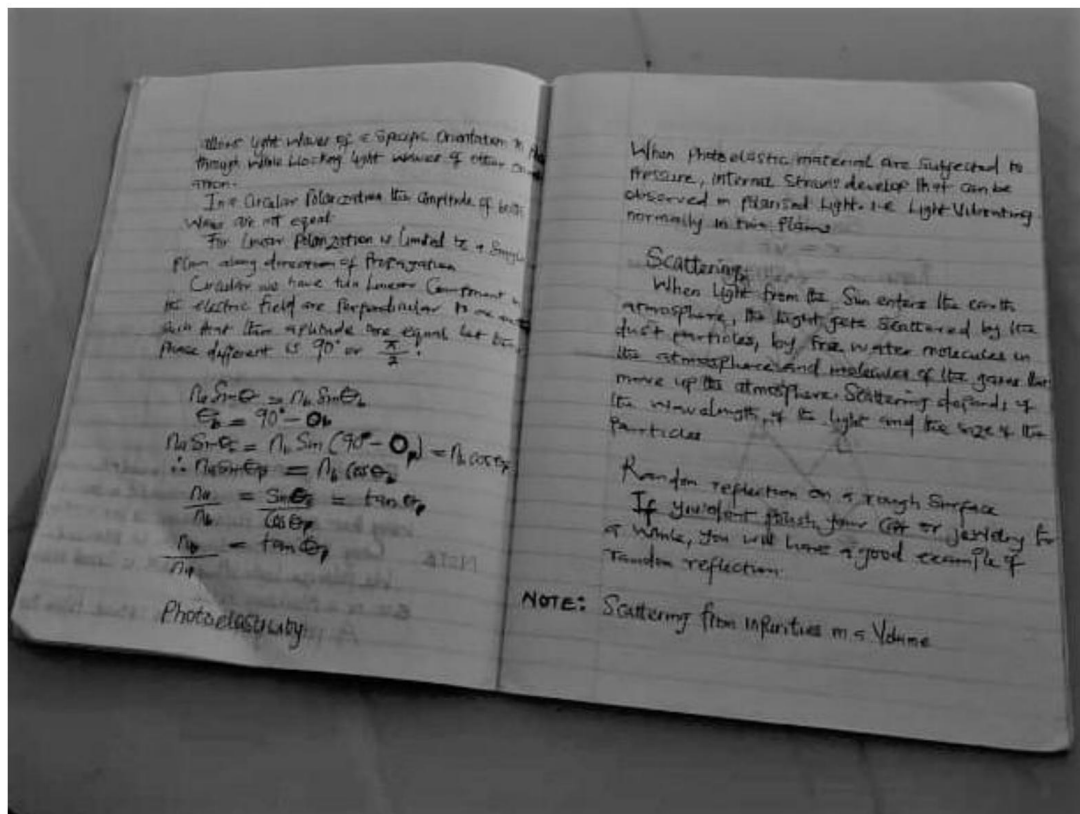
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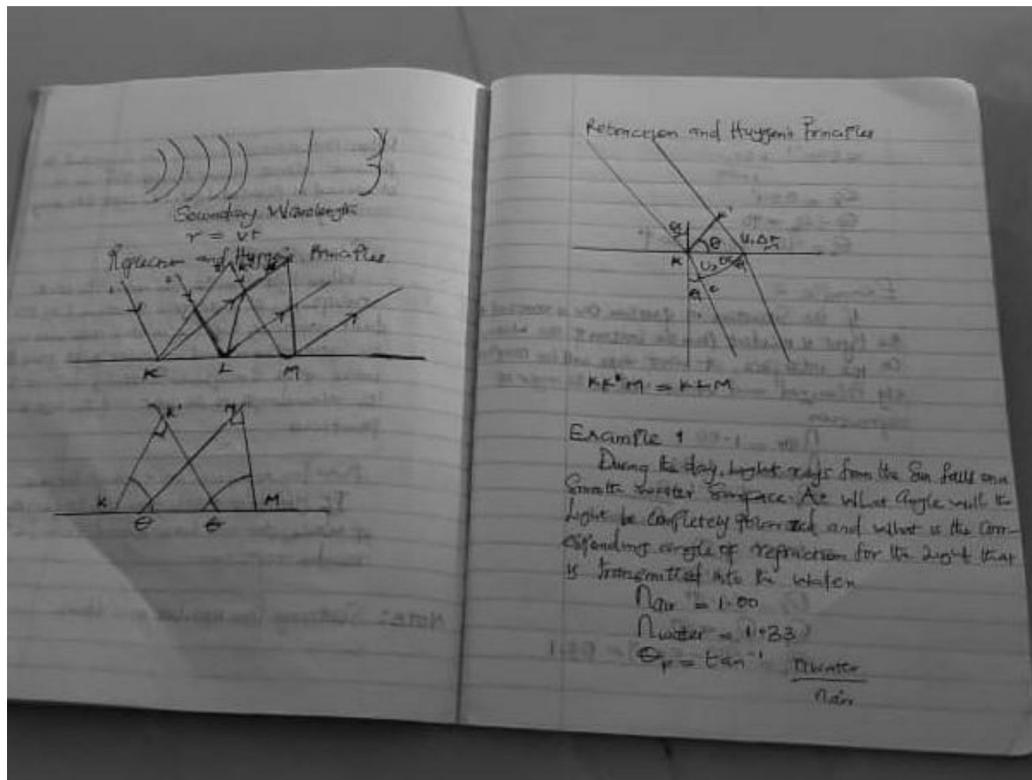
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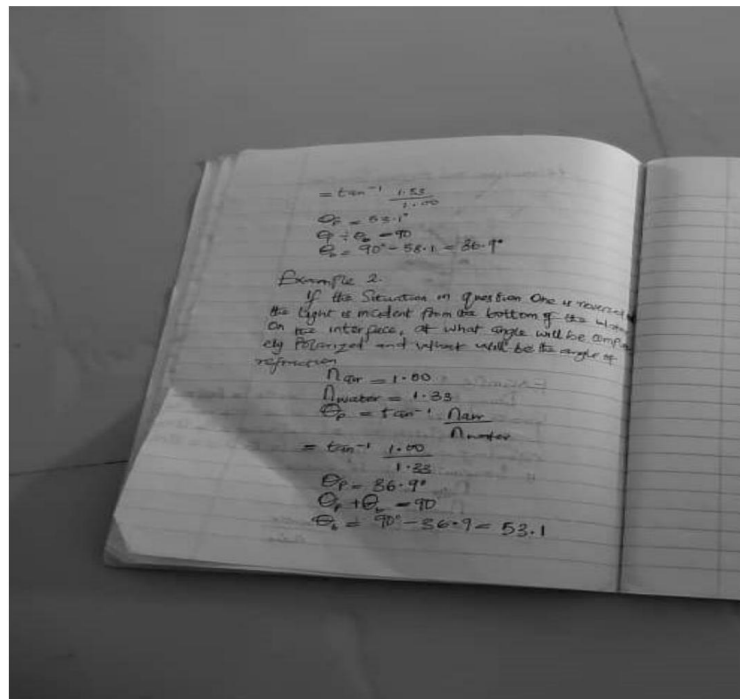


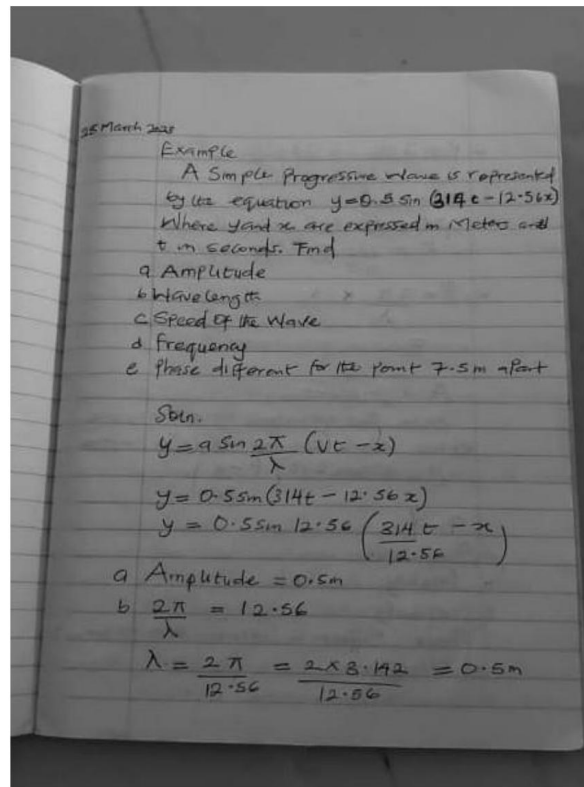


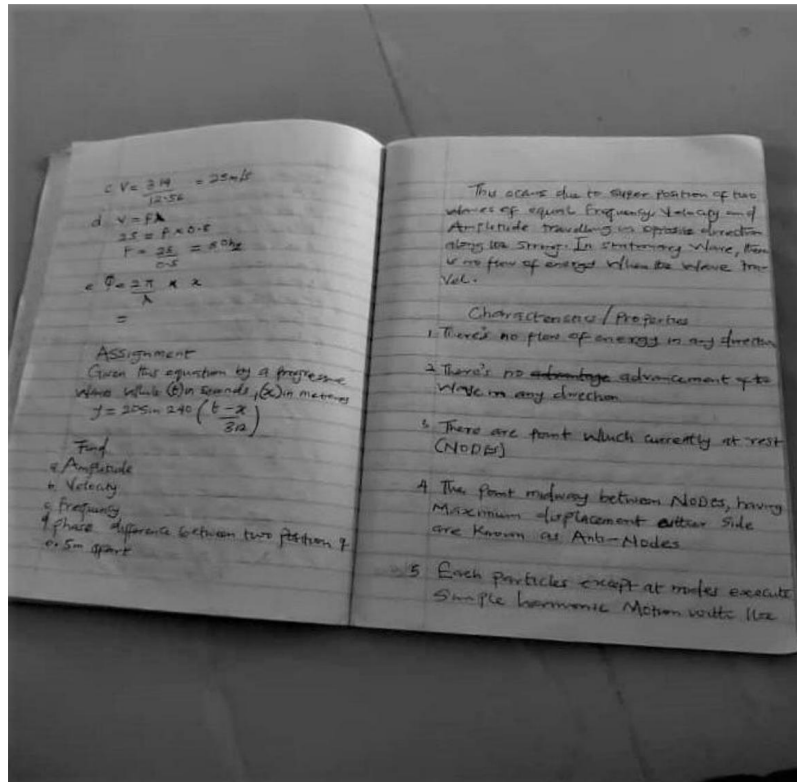


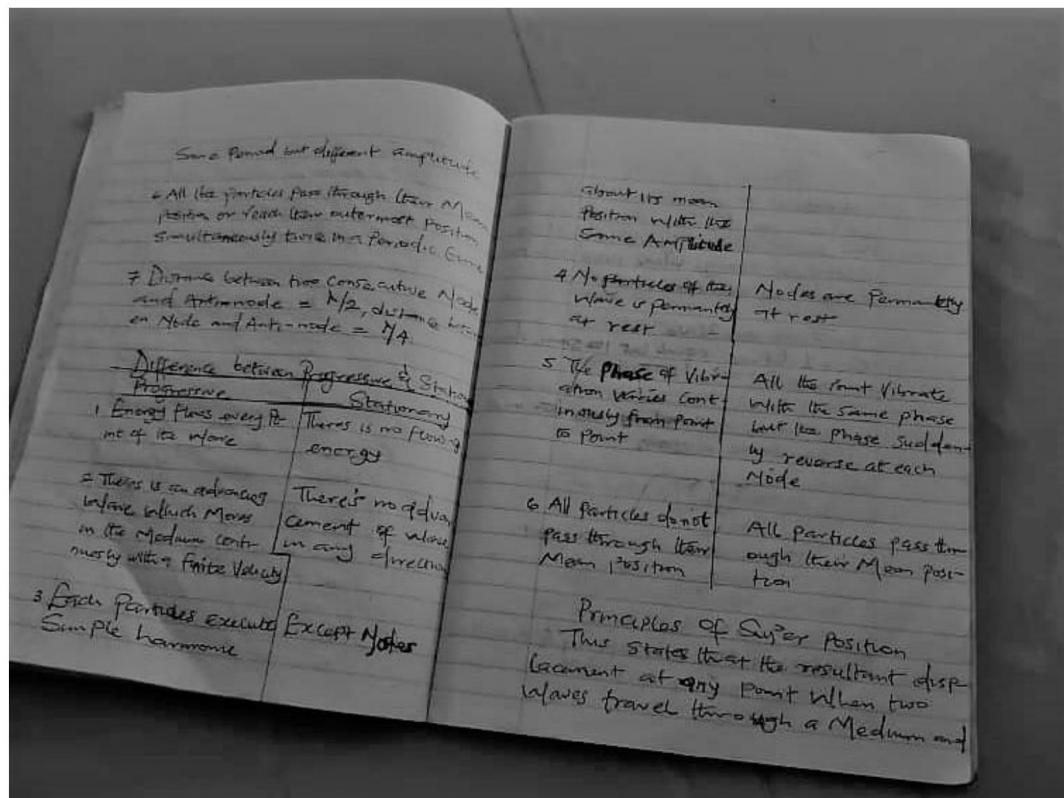








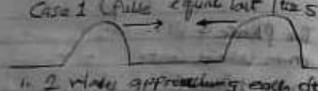




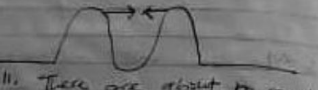
has effect at any point is the sum of the two separate displacements. This principle can be illustrated by means of a long stretched string. Wave pulses are produced at each end simultaneously.

There are 2 cases

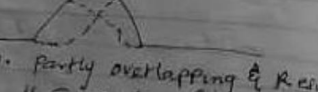
Case 1 (Pulses equal but in same direction)



i. 2 waves approaching each other with the same distance



ii. These are about to meet



iii. Partly overlapping & Resultant is Sum of two Curves

Wave In String.

$$v = \sqrt{\frac{T}{\mu}}$$

If a horizontal rope is fixed at one end and the other end is moved up and down. A wave is created and travels along the rope. The particles of the rope vibrate vertically. The speed ( $v$ ) of the transverse wave depends on the value of Tension ( $T$ ), mass ( $m$ ) and length ( $L$ ) of the rope. It is assumed that the tension ( $T$ ) of the rope has no stiffness. Mathematically it is given as  $v = \sqrt{\frac{T}{\mu}}$  where

$$\mu/L = M \quad \therefore v = \sqrt{\frac{T}{M}}$$

Example 1

Calculate the length of a rope 2.5 kg, when it vibrates the tension in the rope is 26 N and the speed of the particles during the vibration is 12.5 m/s.

Soln.

